ACTION ALIASING IN ROBOT FOUNDATION MODELS

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ABSTRACT

Robot foundation models increasingly expose robot control through language-like or discretized action tokens. This interface assumes that a token preserves the motor distinctions that matter physically. We study the failure case where it does not. We define an action token as a fiber of motor commands and call a token aliased when commands in the same fiber induce different next-state effects under the local dynamics. A simple proposition shows that no decoder that observes only the token can recover distinctions whose effect diameter has already been collapsed. We then introduce Effect-Separating Action Charts (ESAC): audit each token by the diameter of its observed transition effects and split only aliased token fibers into effect charts. In a controlled contact-control simulator with redundant motor nullspaces, coarse action tokens drop from 0.70 to 0.10 success as alias strength grows, while ESAC remains in the 0.66–0.98 range when alias-resolving context is observable. The evidence is deliberately mechanism-level and synthetic; the paper's claim is not that a new foundation model is state of the art, but that action-token interfaces need physical effect audits before their tokens can be treated as robot actions.

1 INTRODUCTION

Robot foundation models (RFMs) and vision-language-action (VLA) policies have made it plausible to treat robot control as another sequence modeling problem: images, language, and sometimes proprioception enter a large model, and a robot action leaves it as a discrete token, action chunk, or continuous policy head (Brohan et al., 2023b;a; O'Neill et al., 2023; Kim et al., 2024; Octo Model Team et al., 2024; Black et al., 2024). This is a powerful abstraction. It is also a physically dangerous one. Language-like action tokens do not inherit the identity conditions of motor commands merely because they are useful for prediction.

This paper asks a narrow question: when does an action token hide multiple physically different commands? We call this *action aliasing*. The issue is not that a policy is uncertain, undertrained, or missing a larger backbone. The issue is representational: after a many-to-one action map has collapsed two commands into one token, a downstream decoder can only choose one effect for the whole fiber. If contact, calibration, embodiment, or force mode makes the commands diverge physically, token accuracy can look high while control is non-identifiable.

Our contribution is intentionally small but sharp.

1. We define token-fiber aliasing using a task-relevant next-state effect metric, rather than raw action distance or language similarity.
2. We prove a lower bound: the worst-case one-step effect error of any token-only decoder is at least half the effect diameter of the token fiber.
3. We introduce Effect-Separating Action Charts (ESAC), which splits aliased token fibers by observed transition effects and trains a chart predictor for decoding.
4. We provide runnable evidence in a controlled embodied-control simulator and release the literature matrix, synthesis docs, and scripts.

We do not claim that all RFMs suffer this failure, that clustering is new by itself, or that ESAC is validated on real robots. The claim is conditional: when a token fiber has large physical effect diameter, token-level success metrics are insufficient, and effect-separating repair changes the central mechanism from *better token prediction* to *preserving physical distinguishability*.

2 Related Work

Robot foundation models and VLA policies. RT-1 and RT-2 showed that transformer policies can map large-scale robot data and web-scale vision-language knowledge into robot actions (Brohan et al., 2023b;a). Open X-Embodiment and RT-X expanded this premise across embodiments (O'Neill et al., 2023); Octo and OpenVLA made generalist policies and VLA models more accessible (Octo Model Team et al., 2024; Kim et al., 2024); pi0 uses a flow model for general robot control (Black et al., 2024). These works make broad robot-data scaling and VLA action emission clearly prior art. Our focus is the action interface itself: whether one emitted token denotes one physically coherent command.

Action tokenization and action representations. Recent work explicitly studies action tokens for VLAs, including FAST (Pertsch et al., 2025), keypoint action tokens (Di Palo & Johns, 2024), discrete policies for multi-task manipulation (Wu et al., 2025), and hybrid discrete-continuous action spaces for assembly (Zhang et al., 2022). Action chunking and ACT-style policies represent temporally extended commands (Zhao et al., 2023), while behavior transformers model multimodal actions (Shafiullah et al., 2022). These methods improve action parameterization. ESAC is different in where it places the criterion: two actions are distinct if their *effects* differ under local dynamics, even when their token, language label, or raw action cluster is the same.

Continuous visuomotor policies. Diffusion and flow policies can model multimodal continuous action distributions without committing to a single discrete token (Chi et al., 2024; Black et al., 2024). They are important alternatives and upper baselines, not straw men. Our theorem targets architectures or APIs that do use a many-to-one token interface; continuous heads can still inherit the problem if an upstream representation has already erased the relevant distinction.

Language grounding, affordances, and contact. VIMA, PaLM-E, SayCan, RoboCat, and related language-conditioned robot systems connect instructions or multimodal prompts to actions and skills (Jiang et al., 2022; Driess et al., 2023; Ahn et al., 2022; Bousmalis et al., 2023). Contact-rich robotics has long shown that small differences in force, compliance, and contact mode can change outcomes (Koval et al., 2015). Action aliasing is where these two facts meet: a semantic action name can be too coarse for the physical distinctions that contact makes decisive.

3 Action-Fiber Aliasing

Let $s \in \mathcal{S}$ be a robot/environment state and $a \in \mathcal{A}$ a motor command. Let $f(s, a)$ denote the one-step dynamics and let $\mathcal{E}_s(a) = \phi(f(s, a)) - \phi(s)$ be a task-relevant physical effect, such as object displacement, contact wrench change, or a learned state-difference embedding. The effect space has metric $d_{\mathcal{E}}$.

An action-token interface is a map $q_s : \mathcal{A} \to \mathcal{Z}$. The state dependence allows tokenizers that include controller mode or embodiment metadata, though many deployed interfaces are closer to $q(a)$.

Definition 1 (Token fiber). *For state s and token z, the action fiber is*

$$F_s(z) = \{a \in \mathcal{A} : q_s(a) = z\}.$$

Definition 2 (Effect diameter and aliasing). *The effect diameter of token z at state s is*

$$\Delta_s(z) = \sup_{a, a' \in F_s(z)} d_{\mathcal{E}}\big(\mathcal{E}_s(a), \mathcal{E}_s(a')\big).$$

The token is ϵ-aliased when $\Delta_s(z) > \epsilon$ for the task-relevant tolerance ϵ.

Proposition 1 (Irreducible token-fiber error). *Fix s and z. Any decoder that observes only z and predicts a single effect $\hat{e}(z)$ has worst-case error*

$$\sup_{a \in F_s(z)} d_{\mathcal{E}}(\mathcal{E}_s(a), \hat{e}(z)) \geq \frac{1}{2} \Delta_s(z).$$

If two actions in the fiber have Euclidean effects separated by $\Delta_s(z)$ and are equally likely, the minimum expected squared effect error is at least $\Delta_s(z)^2/4$.

Table 1: Synthetic contact-control results at alias strength 1.25.

Method	Success	Effect RMSE	Action RMSE
Coarse token prototype	0.148	0.653	1.040
Raw action charts	0.631	0.380	0.966
ESAC, hidden alias context	0.374	0.856	1.171
ESAC, full context	0.743	0.317	0.868
Continuous regression upper	1.000	0.119	0.846
Expert replay	0.999	0.121	0.000

Proof. Choose $a, a' \in F_s(z)$ with effect distance arbitrarily close to $\Delta_s(z)$. By the triangle inequality, $d_{\mathcal{E}}(\mathcal{E}_s(a), \mathcal{E}_s(a')) \leq d_{\mathcal{E}}(\mathcal{E}_s(a), \hat{e}) + d_{\mathcal{E}}(\hat{e}, \mathcal{E}_s(a'))$. At least one term is at least half their separation. Taking the supremum gives the first claim. In Euclidean space, the best single prediction for two equiprobable endpoints is their midpoint, yielding squared error $(\Delta_s(z)/2)^2$ for both endpoints. $\square$

The proposition is elementary, but it changes what should be reported. Token accuracy, language grounding accuracy, and decoder confidence do not certify that a token is physically coherent. The missing quantity is $\Delta_s(z)$.

4 Effect-Separating Action Charts

ESAC is an audit-and-repair layer for tokenized robot policies. Given transitions (s_i, a_i, s'_i, z_i), compute effect vectors $e_i = \phi(s'_i) - \phi(s_i)$. For each token z, estimate an empirical effect radius

$$\widehat{R}(z) = \left(\frac{1}{|I_z|} \sum_{i \in I_z} \|e_i - \bar{e}_z\|_2^2 \right)^{1/2}, \quad I_z = \{i : z_i = z\}.$$

Tokens with small $\widehat{R}(z)$ are left alone. Tokens with large $\widehat{R}(z)$ are split into charts by clustering or covering the effects $\{e_i : i \in I_z\}$. A chart predictor $h(o_i, z_i)$ is trained from observations o_i to chart id, and a chart decoder emits either a prototype action or a local continuous decoder for (z, k).

The important design choice is the clustering space. ESAC clusters within a token by *transition effect*, not by raw motor command. This matters for redundant robots: many motor vectors may be nullspace-equivalent, while nearby motor vectors can have different object effects under contact. ESAC therefore uses the physical effect metric as the action semantics.

5 Controlled Evidence

Simulator. We build a small contact-control environment with state-dependent linear local dynamics $e = B(s)a$. Commands $a \in \mathbb{R}^4$ contain planar motion plus two hidden wrist/force channels. The language-like token keeps only an eight-way cardinal motion label. A binary contact mode and friction-dependent gain determine whether hidden channels create left or right object displacement. Thus two commands can share the same cardinal token while inducing different object effects. We also add random motor nullspace variation: different low-level commands can produce the same object effect, so raw action clustering can waste charts on physically irrelevant differences.

We compare: (i) one prototype per coarse token, (ii) raw action charts that split each token by motor-command clusters, (iii) ESAC with the alias-resolving mode/contact context hidden, (iv) ESAC with full observable context, (v) a continuous regressor as an upper baseline that bypasses token repair, and (vi) expert replay. Success means one-step effect error below 0.36.

Results. Table 1 shows the main operating point. Coarse token decoding fails because the same token needs incompatible effects. Raw action charts help, but ESAC is stronger because its charts ignore nullspace motor variation and split by what the object actually does. The hidden-context ablation is crucial: ESAC is not magic. If observations do not reveal which member of the fiber is

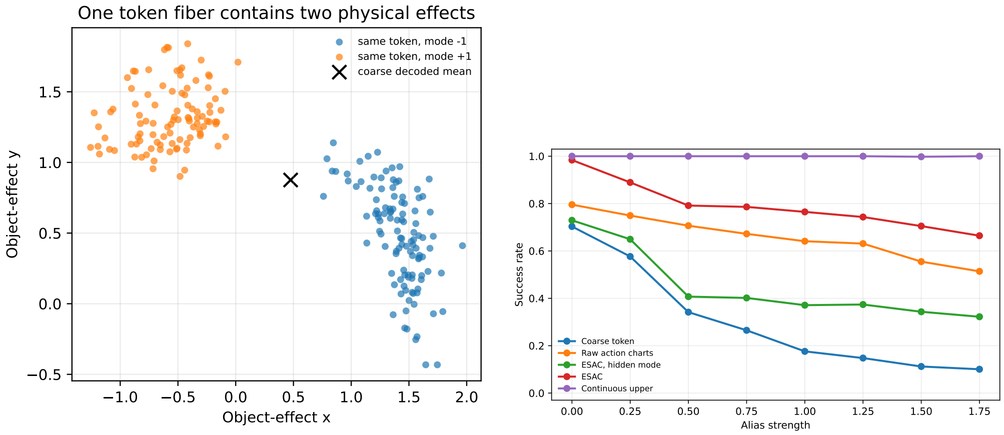

Figure 1: Left: one coarse token fiber contains two physically separated effect modes; the coarse decoded mean lies between them. Right: as alias strength grows, the one-prototype token decoder fails. ESAC remains useful when the alias-resolving context is observable and fails substantially when that context is hidden.

needed, the chart predictor cannot identify it. Figure 1 shows the same pattern across alias strengths; within-token effect radius tracks coarse-token failure in the saved audit plots.

6 LIMITATIONS

This is not a real-robot result. The evidence is a controlled simulator meant to isolate one mechanism. The literature sweep used metadata and abstracts for scale, not full-paper reading for all 3,106 entries. ESAC also requires transition effects during auditing or training, and it requires observations that contain the variables needed to pick the correct chart at test time. Continuous-action policies can avoid this exact token bottleneck, although they may still inherit aliasing from upstream abstractions. Finally, the effect metric ϕ is task-dependent; a bad metric can hide the same problem ESAC is supposed to expose.

7 CONCLUSION

Robot action tokens should not be treated as physically atomic merely because a foundation model can predict them. The right question is whether the token's fiber has small next-state effect diameter under the local dynamics. If it does not, the token is non-identifiable for control. ESAC is a first repair: split action tokens by physical effect charts, not by language names or raw motor proximity. The result is a modest mechanism-level paper, but one with a clear demand for future RFMs: report and repair action-fiber aliasing.

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A REPRODUCIBILITY NOTES

The repository includes `scripts/build_literature.py`, `scripts/synthesize_literature.py`, and `scripts/run_experiments.py`. The experiment command is:

```
python scripts/run_experiments.py
```

It writes `results/sweep_results.csv`, `results/experiment_summary.md`, and the figures included in the paper.

B NOVELTY AUDIT PROTOCOL

Before choosing ESAC, the run built a 3,106-row landscape matrix, a 300-paper serious skim, a 250-paper extraction table, and a 100-paper hostile prior-work set. The nearest hostile areas were VLA models, action tokenization, action chunking, diffusion/flow policies, multimodal prompt policies, and cross-embodiment robot data. These make scale, tokenization, and generic language-conditioned control non-novel. The remaining boundary is the physical effect diameter of action-token fibers.